



Enrichment Exercise # 1

Pre-requisite reading:

http://pages.stern.nyu.edu/~adamodar/New_Home_Page/PVPrimer/pvprimer.htm

A company borrows \$100,000 (the principle) today and promises to pay 5% interest each year and then repay the principle in 5 years. Calculate the present value of this future stream of payments. Hint: The interest payments represent an annuity, the principle is a one-time payment, the number of periods is 5 and the rate is 5%.

You will need to calculate the present value of the 2 components, the principle repayment in the future as well as the periodic interest payments:

COMPONENT 1: Present Value of the \$100,000 =

$$\frac{\$100,000}{1.05^5} = \$78,352.62$$

COMPONENT 2: Present Value of the Annuity (\$5,000 a year for 5 years) =

$$\$5000 \left[\frac{1-(1+.05)^{-5}}{.05} \right] = \$21,647.38$$

Total Present Value of the 2 components = 78,352.62 + 21,647.38 = \$100,000

When you calculate the future components back to the present, you are discounting (or stripping away) the future interest. Thus, the present value is the amount borrowed.